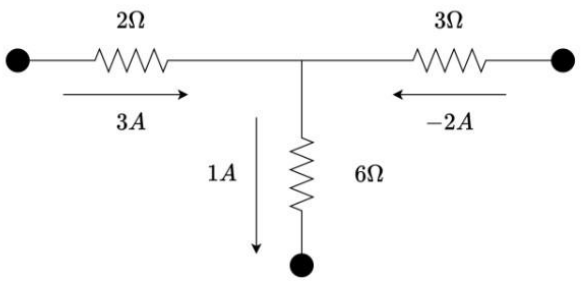
ASSIGNMENT-1

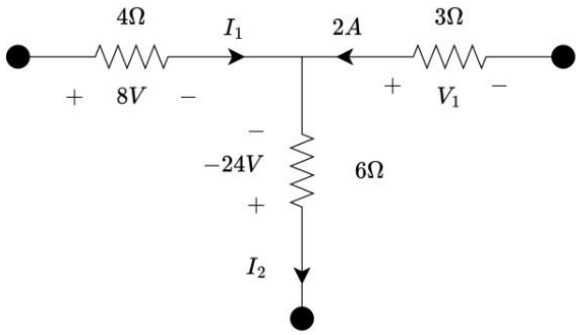
Course: CSE250

Total Marks: 50

Q1. [CO1] 10



(a)

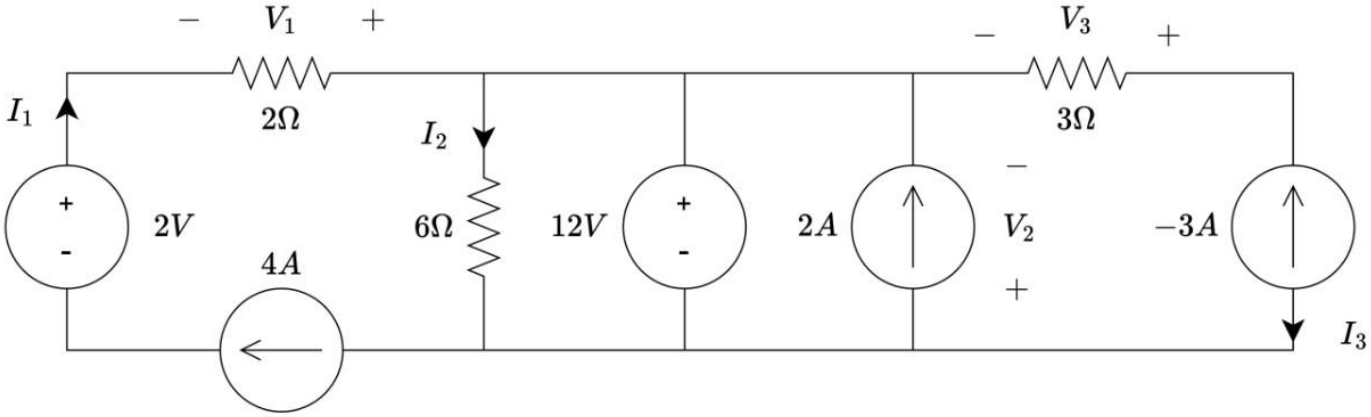


(b)

a) Redraw the circuit (a) with the voltage associated with the resistors. Marks the polarity of the voltages.

b) Find the value of V_1, I_1 and I_2 in circuit (b).

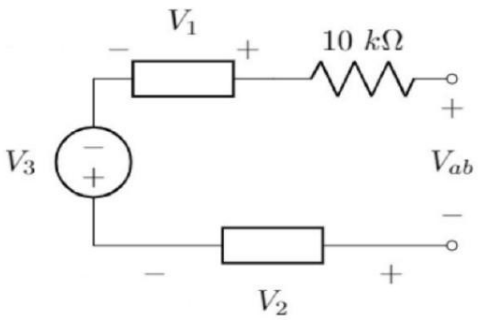
Q2. [CO1] 10



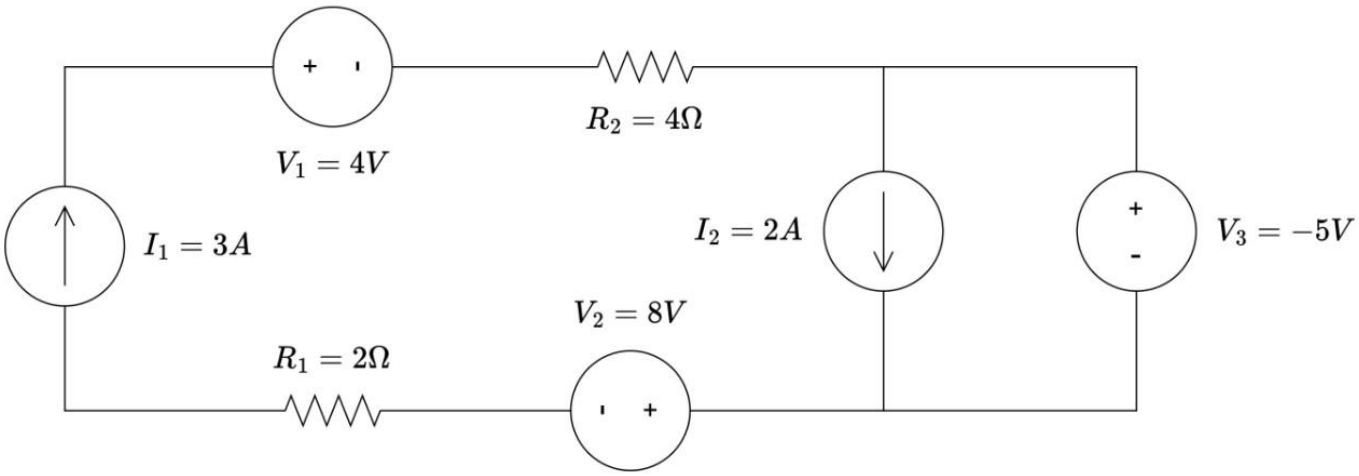
Find the Values of I_1, I_2, I_3, V_1, V_2 and V_3 from the circuit

Q3. [CO1] 5

Find V_{ab} when V_1 = 100 V, V_2 = 60 V and V_3 = 50V. What is the voltage drop across 10 KΩ and why?

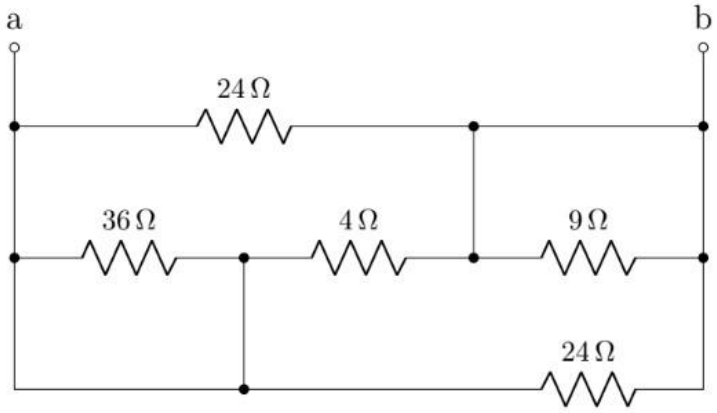


Q4. [CO1] 10



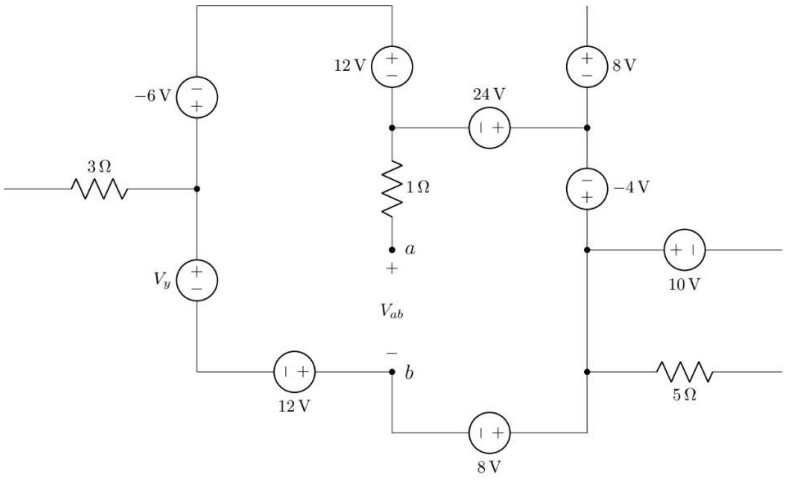
Find the power associated with R_1, R_2, V_1, V_2 and V_3. Also mention whether these devices are supplying or consuming power.

Q5. [CO1] 5



Find R_{ab}, the equivalent resistance between the terminals a and b from the circuit.

Q6. [CO1] 10



a) Identify the loop and determine the value of V_y using KVL.

b) What is the voltage V_{ab} across the open terminals a and b.